

Mooketsi Kosa

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Professional Summary

Data Engineer and Software Developer with hands-on experience designing and building data pipelines, RESTful APIs, and backend systems. Pursuing a BSc in Computer Science and Informatics at the University of Johannesburg. Proficient in Python, SQL, Apache Kafka, PostgreSQL, and Cassandra, with practical experience in real-time data ingestion, stream processing, and database design. Passionate about transforming raw data into reliable, queryable infrastructure that drives decision-making.

Technical Skills

- Languages:** Python, SQL, Java, JavaScript, C++, C#, Visual Basic
- Data & Streaming:** Apache Kafka, Data Pipelines, ETL/ELT, Real-Time Stream Processing, Batch Processing
- Databases:** PostgreSQL, MySQL, Cassandra (NoSQL), Database Design, Query Optimisation
- Frameworks & APIs:** FastAPI, Flask, Quarkus, Node.js
- Tools & Platforms:** Git, GitHub, Visual Studio, Google Analytics, Excel
- Concepts:** Distributed Systems, Data Modelling, Schema Design, OOP, Agile, RESTful APIs

Education

- University of Johannesburg** Feb. 2022 – Nov. 2026 (Expected)
Bachelor of Science in Computer Science and Informatics Johannesburg, Gauteng
- Relevant Coursework:** Data Structures, Algorithms, Databases, Computer Systems, Machine Learning
 - Certificate in 4IR (Fourth Industrial Revolution); Certificate in African Insight
- Gritlab Africa** 2023
Certificate of Completion – Introduction to Python (3-Month Course) Online
- Completed a structured 3-month programme covering Python fundamentals, algorithms, and data-oriented problem-solving.

Experience

- Johannesburg Business School (JBS)** Aug. 2023 – Nov. 2025
Software Developer Intern Johannesburg, Gauteng
- Analysed system requirements and implemented targeted data integration improvements, reducing manual data handling across multiple platforms.
 - Collaborated with cross-functional teams in an agile environment to deliver data-driven software solutions addressing real-world business problems.

Projects

- FarmSmart Analytics Platform** | Python, Flask, PostgreSQL, REST API 2026
- Built the data layer for a farm management platform, designing a PostgreSQL schema to store and query crop, livestock, and expense records, with a Flask REST API serving analytics to the front-end.
 - Integrated external weather data feeds into the pipeline to enrich farm records and power real-time dashboard analytics.
- RESTful API Service** | Java, Quarkus, PostgreSQL 2025
- Designed a scalable RESTful API using Quarkus and Java with PostgreSQL as the primary data store, implementing authentication, input validation, and unit tests following test-driven development practices.
 - Produced thorough API documentation and applied schema design best practices to ensure data integrity and query performance.